RITIKA SINGH

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SUMMARY

Pre-final year B.Tech Computer Science Engineering (Cyber Security) student and full-stack developer who builds and deploys secure MERN applications end to end. Shipped two full-stack platforms with JWT authentication, Role-Based Access Control (RBAC), and REST API design, and solved 130+ Data Structures and Algorithms (DSA) problems on LeetCode. Seeking a Software Engineer / Software Developer role, applying cybersecurity as a technical differentiator.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL
Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap
Backend: Node.js, Express.js, REST APIs
Databases: MongoDB, MySQL
Security: JWT Authentication, Role-Based Access Control (RBAC), Input Validation, CORS, Helmet, Rate Limiting
Core CS: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
Tools: Git, GitHub, Postman, VS Code, Vercel, Render

PROJECTS

AI CyberTrustShield | *React,Node.js,Express.js,MongoDB,JWT,OCR,PDFParser* — GitHub

- Architected a full-stack MERN cybersecurity platform flagging fake job offers, phishing emails, fraudulent websites, and scam PDFs via rule-based content and pattern analysis.
- Implemented JWT authentication and Role-Based Access Control, securing access across user roles.
- Integrated OCR and PDF parsing to extract and verify text from scanned documents and reports.
- Engineered REST APIs for scam detection, report generation, and management; hardened backend with Helmet, rate limiting, input validation, and CORS.
- Deployed frontend on Vercel and backend on Render with production configuration.

Password Security Analyzer | *React, Node.js, Express.js, MongoDB, JWT, REST API* — GitHub

- Built a full-stack MERN platform for real-time password strength evaluation using entropy calculation, pattern detection, and dictionary-based checks.
- Implemented JWT authentication with secure password hashing, storing analysis history in MongoDB.
- Designed REST APIs for analysis, authentication, and user management, paired with a responsive React interface delivering real-time feedback.
- Secured backend with rate limiting, Helmet, CORS, and input validation; deployed to Vercel (frontend) and Render (backend).

INTERNSHIP EXPERIENCE

AndroidDeveloper—VirtualInternship,EduSkills | *NationalInternshipPortal,inassociationwithGoogleforDevelopers'IndiaEDUProgram* | *Jan–Mar 2025*

- Built and debugged Android application modules, applying core mobile development concepts including app structure, UI development, and component-based design.
- Applied software development practices and object-oriented design patterns throughout the build and improvement of Android-based applications.
- Worked independently in a remote setting, managing the full development-to-debugging workflow for assigned modules.

EDUCATION

IES College of Technology, Bhopal (RGPV University)
B.Tech, Computer Science Engineering — Cyber Security Specialization | CGPA: 8.56/10.0 | 3rd Year

ACHIEVEMENTS

- Winner, 2 hackathons; participant in 5+ hackathons overall — contributed to end-to-end project development and presentation.
- Smart India Hackathon (SIH) — College-Level Finalist; Grand Finale Waitlist Finalist.
- College Topper — Academic Excellence Recognition, 6 times (CGPA 8.56/10.0), with cash prizes for outstanding academic performance.
- National-Level Ideathon Finalist; recognized participant at Microsoft Build Local Host, Bhopal.
- Solved 130+ Data Structures & Algorithms problems on LeetCode (Arrays, Strings, Trees, Dynamic Programming, Graphs).

CERTIFICATIONS & TRAINING

- Full Stack Web Developer Bootcamp — Udemy • Generative AI Camp — Algo University
- B10X Digital Workshop — AI Tools & Technologies: hands-on exposure to generative AI and AI-assisted productivity tools
- NPTEL — Ethical Hacking & IoT (Penetration Testing, Vulnerability Analysis, Exploit Development) • NPTEL — DBMS (Advanced SQL, Normalization, Indexing)
- Power BI (Micro-course) • Microsoft Excel — Beginner & Advanced Level